

James George

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SUMMARY

Senior Data Engineer with **10+** years of experience designing and scaling data platforms across **AWS** and **GCP**. Proven expertise in building **lakehouse architectures**, **real-time streaming pipelines**, and **dimensional warehouses**, enabling secure, compliant, and cost-efficient analytics for **Fintech**, **e-Commerce**, and **Healthcare** domains. Skilled in **Spark**, **Flink**, **Airflow**, **dbt**, **Redshift**, **Snowflake**, and **Delta Lake**, with a strong track record of modernizing legacy systems, optimizing query performance, and deploying ML models via **SageMaker** and **Vertex AI**. Adept at collaborating with cross-functional teams, driving governance frameworks, and delivering measurable business impact through scalable, production-ready data solutions.

TECHNICAL SKILLS

- **Languages:** Python, Go, SQL, Scala, JavaScript
- **Backend & APIs:** FastAPI, Gin, gRPC, REST
- **Data Engineering & Pipelines:** Apache Spark, Flink, Airflow, Dagster, ETL/ELT, Delta Lake
- **Cloud Platforms:** AWS (S3, Redshift, EMR, Lambda) , Azure (Data Factory, Synapse, Blob Storage, Event Hubs),
- **Data Engineering:** dbt, Airflow, Delta Lake, Redshift, Snowflake, BigQuery, Data Lakes
- **CI/CD & Infrastructure:** GitHub Actions, Terraform, Jenkins
- **Security & Auth:** OAuth2, JWT
- **Data Visualization & Monitoring:** Matplotlib, Tableau, Power BI, Grafana, Prometheus

WORK EXPERIENCE

Senior Data Engineer | ICF | Chicago, IL | 09/2021 – Present

- Led architecture design for enterprise-scale **data lakes** on **AWS S3** with **Delta Lake**, enabling unified storage for **Fintech** datasets.
- Architected event-driven pipelines using **AWS Kinesis**, **Lambda**, and **Step Functions**, supporting real-time fraud alerts and personalized shopping experiences.
- Architected migration strategy from legacy **ETL systems** to **AWS EMR** and **Glue**, improving scalability and reducing operational overhead.
- Defined data **governance frameworks** with **AWS Lake Formation** and **Glue Catalog**, ensuring compliance with **PCI DSS** and **GDPR** across **Fintech**.
- Partnered with product and data science teams to design **ML feature stores** on **Redshift** and **Snowflake**, accelerating fraud detection and recommendation models.

- Designed and implemented a scalable **Big Data Lakehouse architecture** on **AWS S3** and **Delta Lake**, structured with **Medallion principles**, reducing data duplication by 40% and improving governed access.
- Modernized enterprise **data platforms** by migrating fragmented reporting systems into dimensional - modeled **Redshift warehouses**, significantly enhancing query performance and data consistency.
- Designed scalable ETL pipelines using **PySpark** on **AWS EMR** and **Databricks**, processing multi-terabyte datasets with optimized partitioning, broadcast joins, and predicate pushdown.
- Mentored **6 junior engineers** on **Spark optimization**, **Airflow DAG design**, and **AWS best practices**, accelerating team velocity and improving code quality.
- Engineered **real-time pipelines** with **AWS Kinesis**, **Lambda**, and **Step Functions**, enabling instant fraud alerts in **Fintech**.
- Migrated batch workflows to **AWS Lambda**, eliminating server overhead, reducing operational costs by 40%, and improving scalability under variable workloads.
- Developed **scalable data pipelines** on **Databricks**, leveraging **Apache Spark** for batch and streaming workloads, which reduced processing time for multi-terabyte datasets by 25% and improved reliability of downstream analytics.
- Optimized **Redshift** workloads through advanced partitioning, distribution keys, and workload management, cutting query runtimes by 35% and lowering compute costs.
- Implemented **CI/CD pipelines** for data workflows using **GitHub Actions**, **Terraform**, and **Jenkins**, enabling automated testing, deployment, and rollback across **AWS** environments.
- Built observability dashboards with **CloudWatch**, **Grafana**, tracking pipeline health, SLA adherence, and data freshness across mission-critical workloads.
- Collaborated with data science teams to deploy **ML models** via **AWS SageMaker**, enabling scalable fraud detection and recommendation systems in production.
- Piloted adoption of **Databricks Lakehouse architecture**, standardizing ingestion and transformation patterns across teams and enabling unified **governance**, which decreased data quality issues by 30% and accelerated delivery of business insights.
- Delivered executive-facing analytics layers in **Tableau/Power BI**, surfacing KPIs for fraud detection, patient outcomes, and e-Commerce revenue growth, directly influencing leadership decisions.
- Partnered with **DevOps teams** to integrate **IAM policies**, **secrets management**, and **Terraform templates**, strengthening security posture and standardizing infrastructure-as-code practices.
- Delivered **Fintech project** consolidating multi-source transaction data into a unified **AWS data lake**, improving fraud detection accuracy and regulatory reporting.
- Completed **e-Commerce project** integrating customer, product, and order pipelines, enabling advanced personalization, inventory optimization, and revenue growth analytics.
- Automated pipeline and deployment alerts into **Slack channels**, improving visibility and reducing incident response times.

Data Engineer | Optum | San Francisco, CA | 11/2019 – 08/2021

- Engineered and optimized large-scale ETL/ELT pipelines using **Apache Spark** and **Flink** on **Azure Databricks** and **Azure Data Factory**, achieving a **40% reduction in latency** for healthcare data ingestion.
- Developed and deployed backend **APIs** with **FastAPI**, **Gin**, and **gRPC**, integrating with **Cloud Endpoints** to securely expose patient data services across multiple **healthcare** applications.

- Implemented Delta Lake architecture on **Azure Data Lake Storage (ADLS)**, ensuring ACID transactions and reliable ingestion of electronic medical records (EMR) to support HIPAA-aligned data governance.
- Enhanced data reliability by implementing **data quality frameworks** with anomaly detection, validation rules, and automated reconciliation scripts in **Python/SQL**, building trust in enterprise datasets.
- Designed dimensional and semantic **data models** with **dbt** and **SQL** for **BigQuery** and **Snowflake**, standardizing analytics across clinical research, claims, and operational datasets.
- Automated complex **workflows** with **Airflow** and **Dagster**, orchestrating batch and streaming pipelines that processed clinical trial data and **IoT telemetry** from medical devices.
- Leveraged **Azure Event Hubs** and **Azure Stream Analytics** to build real-time ingestion pipelines for patient monitoring systems, enabling near-instant alerts for critical **health** events.
- Engineered **ML-ready datasets** by combining **Python** and **Scala** transformations, supporting predictive modeling initiatives such as patient risk scoring and hospital readmission forecasting.
- Established **CI/CD pipelines** with **GitHub Actions**, **Terraform**, and **Jenkins**, automating infrastructure provisioning and reducing deployment cycles from weeks to days.
- Integrated **OAuth2** and **JWT authentication** into microservices, ensuring secure and role-based access to sensitive healthcare datasets across distributed systems.
- Implemented proactive **monitoring** with **Grafana** and **Prometheus**, creating custom dashboards to track pipeline health and reducing incident response times by 25%.
- Built interactive **dashboards** in **Tableau** and **Power BI**, enabling clinicians and administrators to visualize patient outcomes, operational KPIs, and compliance metrics.
- Developed modular data models with **dbt**, **Snowflake**, and **BigQuery**, standardizing transformations and improving query performance across business domains.
- Optimized **BigQuery queries** and **partitioning strategies**, cutting reporting runtimes from hours to minutes and lowering query costs by 30% for healthcare analytics teams.
- Delivered a major healthcare data platform initiative integrating EMR, claims, and IoT data streams into a unified **Azure-based data lake**, improving patient-care insights and regulatory compliance reporting.
- Documented **workflows, standards, and onboarding guides**, ensuring knowledge transfer and consistency across engineering teams working on healthcare and enterprise data projects.
- Collaborated with cross-functional stakeholders (product, compliance, finance) to align data engineering deliverables with business priorities, ensuring analytics directly supported strategic decision-making.

Data Engineer | Planet | San Francisco, CA | 05/2018 – 10/2019

- Engineered and optimized **ETL/ELT pipelines** using **Apache Spark**, and **Delta Lake**, enabling scalable ingestion and transformation of multi-terabyte datasets.
- Implemented real-time streaming solutions with **Apache Flink** and **Kafka**, reducing latency in fraud detection and recommendation workflows.
- Developed reusable **PySpark** libraries for schema enforcement, data quality validation, and metadata-driven transformations, improving pipeline consistency across teams.
- Built cloud-native data platforms leveraging **AWS S3**, **Redshift**, **EMR**, **Lambda** and **Dataflow**, **Pub/Sub**, ensuring cost-efficient storage and analytics.
- Automated CI/CD workflows using **Jenkins**, and **Terraform**, streamlining deployment of data pipelines and infrastructure.
- Worked on **SQS** Queue receiver using **Spark Streaming** context to consume the data from extended queue and integrated with ETL Functions

- Optimized query performance on **Redshift** and **BigQuery** through distribution key tuning, materialized views, and workload management, reducing reporting runtimes by 40%.
- Collaborated with cross-functional teams to deliver secure APIs and dashboards, integrating **OAuth2/JWT authentication**.

Junior Data Engineer | C.H Robinson | Miami, FL | 07/2015 – 04/2018

- Developed **ETL pipelines** in **Python** and **SQL**, automating ingestion from multiple relational databases and log files, which reduced manual data preparation time and ensured consistent delivery of daily datasets.
- Built and maintained **Apache Airflow DAGs** to orchestrate batch workflows, adding retry logic and alerting mechanisms that improved pipeline reliability and reduced job failures by 20%.
- Implemented **Flink streaming applications** to process near-real-time clickstream and transaction data, enabling dashboards with latency under five seconds for operational monitoring.
- Provisioned **AWS infrastructure** with **Terraform**, deploying EMR clusters, S3 buckets, and IAM roles, gaining hands-on experience with Infrastructure-as-Code practices.
- Leveraged **AWS Lambda functions** to trigger lightweight transformations and event-driven workflows, reducing overhead for small batch jobs and simplifying pipeline logic.
- Automated **CI/CD deployments** using **Jenkins**, integrating unit tests and schema checks to ensure safe promotion of pipeline updates into production.
- Optimized **Spark jobs** on AWS EMR for large-scale data transformations, cutting runtime by 25%.
- Built **Tableau** and **Power BI reports** to visualize KPIs, enabling stakeholders to track performance and trends.
- **Documented workflows, pipeline architecture, and best practices**, supporting onboarding and knowledge transfer across the engineering team.
- Collaborated in **Agile/Scrum teams**, participating in sprint planning, daily stand-ups, and retrospectives to deliver incremental improvements in data pipelines and ensure alignment with business priorities.

EDUCATION

University of Florida | Gainesville, FL | 09/2011 – 06/2015

BS, Computer Science